



3D WALLPAPER STOCK MANAGEMENT SYSTEM

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**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2022

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3D WALLPAPER STOCK MANAGEMENT SYSTEM

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NORAPHIT PORNJARASKUNWATHANA
SUPAWISH KODYEE**

**A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

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2022

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BEK SHUANG ZHEN

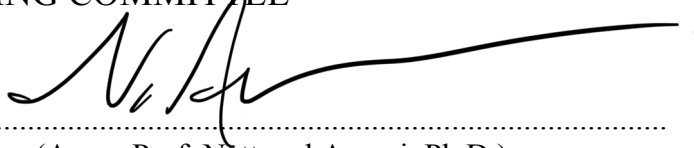
NORAPHIT PORNJARASKUNWATHANA

SUPAWISH KODYEE

THIS SENIOR PROJECT HAS BEEN APPROVED
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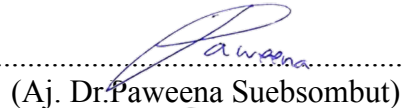
2022

EXAMINING COMMITTEE



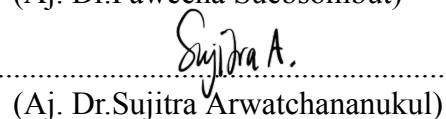
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Abstract

This Project is designed to support multiple forms of stocking products. The project organizers have a desire to bring the knowledge that has been studied, learned and developed to be useful for marketing and can be used in several ways.

Keyword: Web Application, MySQL, Management

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Stock management is a thing that is often overlooked. Even though it is very important to business because it is believed that many business owners may have encountered the problem of insufficient product supply. There are too many types of products or the number of missing products. In which some problems have a considerable effect on profit and loss. Nowadays, stock management is very important, if a business has a product stock system and stock management method, it will help to work more efficiently, for example, helping to keep the products in the stock organized, know which products sell well rarely sold to evaluate or make a decision to stop production, sell or produce more correctly, reducing the chance of error in work. Stock management is an important aspect of doing business. If we have good stock management, it will help reduce costs, save time and also make it easier to check product stock. Even if we have effective stock management it will give our business an advantage over many competitors.

1.2 Objective

- 1.2.1 To continuity in sales
- 1.2.2 To know when to order more
- 1.2.3 To know the current number of products

1.3 Scope

There are two roles that can use system, which is staff and admin

1.3.1 Staff

- Can create, add, update and delete products.
- Login / Logout system.
- View inventories.
- View histories
- Can export Excel files.
- Can edit supplier and customer information

1.3.2 Admin

- Can create, add, update and delete products.
- Login / Logout system.
- View inventories.
- View histories.
- Can export Excel files.
- Can edit supplier and customer information.
- Can create and update user accounts.

1.4 Methodology

1.4.1 Proposal preparation

- We find the topic of the project and discuss it with group members, then we consult it to the advisor and committee. We have studied and analyzed the requirements of the project or proposal, including the scope of work, objectives, and expected outcomes. Additionally, we have clearly defined the objectives of the project or proposal. In designing the proposal, we outline the steps that will be taken in the project, including the timeline and resources required.

1.4.2 Get and prepare requirement

- We find related research to compare with the topic that the group has chosen and analyze their advantages and disadvantages. We have engaged in discussions with all stakeholders or project participants to understand their requirements and expectations. We analyzed the information received from stakeholders to gain insights into the desired patterns. We collected and documented all relevant information related to the project's requirements, encompassing both functional and technical needs.

1.4.3 System design and Develop Application

- Understand the requirements of the system we want to design and develop.
- Research and collecting tools, software that we think is appropriate for the topic.
- Gather information as needed to design the system by analyzing the operation of the system, then the results of the study and analysis are written into a Diagram.

Final Project presentation

- Presentation of the project to the advisor and committee.
- We have presented our project or work, detailing the methodology employed throughout the project. We showcased the results obtained, including crucial data. The

presentation concluded with a summary of the work, emphasizing its significance.

Additionally, we outlined the plans for future steps and how we intend to proceed.

Final Project Document

- Bring the proposal to the advisor and committee for certification and send it to the lms website.
- We have attached details regarding the project, its objectives, and significance. We have outlined the methodology employed in carrying out the project, including the data used, encompassing both initial data and information collected during the project. Results obtained from the project have been presented, including tables or visuals to illustrate the data. A summary of the work has been provided, emphasizing key points. The document also outlines the timeline for the future plans and includes a list of references used in the document.

1.4.4 System Development & Testing

- It is the process of programming (Coding) to develop a system from designed features. Then do a test to find errors (Testing) to check the accuracy. Until we are confident that it is correct and check if there is a problem, If we find that there is an error caused by the operation of the system we will fix it. The system must be corrected and ready for use before being actually used.
- Use the results of the analysis to design a concept. At this stage, we will identify technical system behavior, details of the equipment, technology, design database importing data, Characteristics of the resulting report format and results.
- Install and configure the server and software, write and test the code.
- Repeat the test to make sure the system can be used.
- We continue to fix problems, improve and develop the system.

1.4.5 System Maintenance

- It is a continuous system maintenance procedure after starting operations. Users of the system may encounter problems later. Therefore, we should follow up and evaluate, collect mistakes and analyze them to improve the system.

1.4.6 Final Project 2 Presentation

- Presentation of the project to the advisor and committee. We have followed a similar approach to the presentation in the first term.

1.4.7 Final Project 2 Documentation

- Bring the proposal to the advisor and committee for certification and send it to the lms website. We also follow a similar approach to the documentation in the first term.

1.5 Plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 exam										
System development										
System testing										
Senior project 2 document preparation										
Senior Project 2 exam										

1.6 Expected Result

1.6.1 Staff and Admin can manage products in stock.

1.6.2 Staff and Admin can check product histories.

1.6.3 Admin can create and update user accounts.

1.7 Place and Equipment

1.7.1 Place

- Mae Fah Luang University Library
- Dormitory

1.7.2 Equipment

- Computer and Laptop

1.7.3 Software

- Visual Studio Code (Coding Tool)
- MySQL Database
- Vue.js
- Figma (Design UX/UI)

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

A system design of UX/UI is frequently based on Figma [4] and we include several diagrams such as use case diagram, E-R diagram and workflow diagram to design or create our system function. This system has emerged from various factors, including research conducted by exploring similar systems and analyzing the workflow patterns of those systems. We gather data from various systems for analysis, considering their advantages, disadvantages, and limitations. We then adapt this information to align it as closely as possible with our own system.

2.2 Related Work

We have referenced the existing stock system and adapted it to our stock system by modifying the functionality to be as relevant to our topic as possible. The stock system we refer to is as follows.

1. E count: there are features that can Accounting/Finance, Sales/Trading, Purchasing/Inventory/Logistics, HR/Payroll, Production/Manufacturing/Cost, GW(Groupware)

2. Flow Account: there are features that can Feature for Business Owners, Feature for Accountant, Quotation, Receipt, Billing Notes/Invoices, etc.

From these two brands, we have implemented many features in our system such as Accounting, Sale, Trading and Inventory to create our system Accounting, we will assign administrators or admin to manage all user accounts on the system, admin can edit information and can create new user accounts for users in the system. Production, Sale and inventory, this feature will be importing and exporting products from the warehouse system or managing various warehouses, including editing the product information.

The advantage is that we can use programs and work according to our abilities, maximizing our efficiency. However, the drawback is that there may be many other programs and workflows that are suitable for our tasks, but we may not be utilizing them.

2.3 Related Technology

In terms of determining the technology used. There will be a part of the software that we have defined that we will use Visual Studio code, MySQL Database, Vue.js and Figma in our main work with details as follows.

2.3.1 Visual Studio Code

Visual Studio Code is a lightweight but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support

for JavaScript, TypeScript and Node.js and has a rich ecosystem of extensions for other languages and runtimes is shown in Figure 2.1

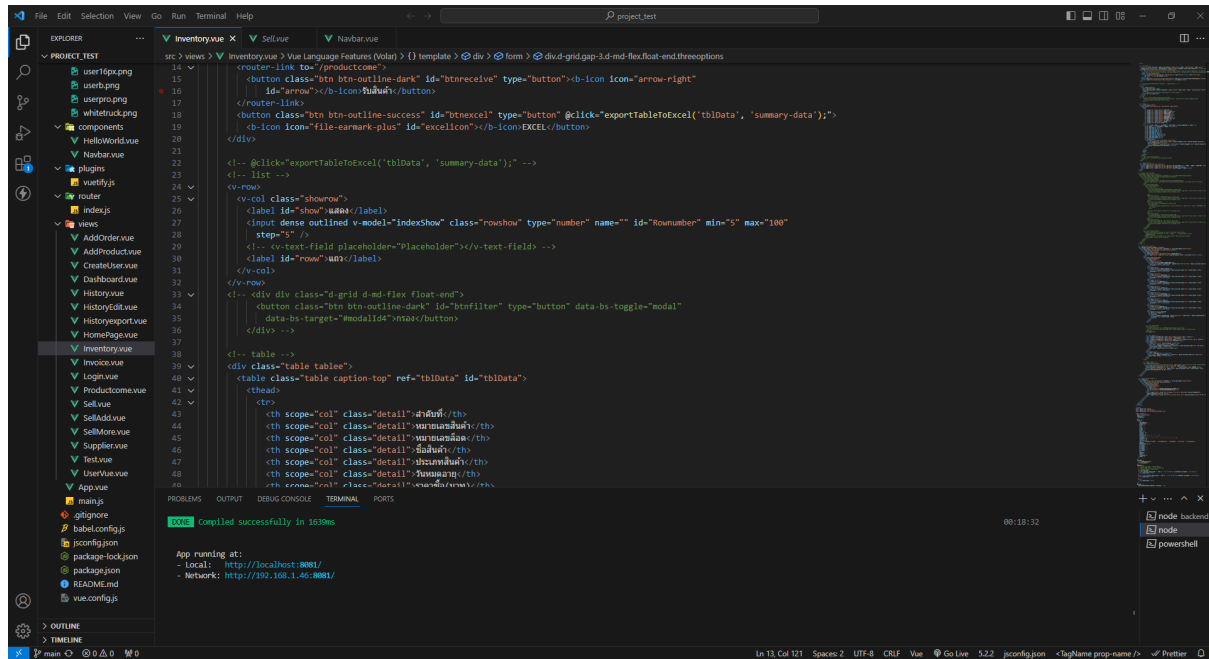


Figure 2.1 Visual Studio Code main page^[1]

2.3.2 MySQL Database

MySQL is the world's most popular open-source database. According to DB-Engines, MySQL ranks as the second-most-popular database, behind Oracle Database. MySQL powers many of the most accessed applications, including Facebook, Twitter, Netflix, Uber, Airbnb, Shopify, and Booking.com. is shown in Figure 2.2.

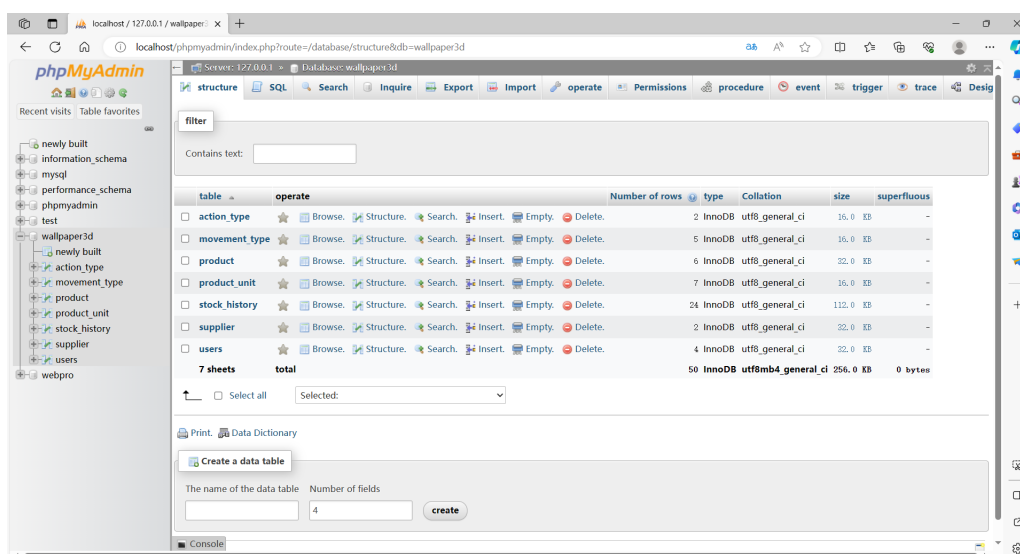


Figure 2.2 MySQL Database^[2]

2.3.3 Vue.js

Vue.js is a dynamic JavaScript framework for building user interfaces. This system is known for its steep learning curve. However, it is a very approachable and straightforward framework. We may start building web apps in Vue.js only with a basic understanding of HTML, CSS, and JavaScript. A short learning curve is a feature of this progressive framework. It is flexible and can be used as a comprehensive utility or framework in web development is shown in Figure 2.3

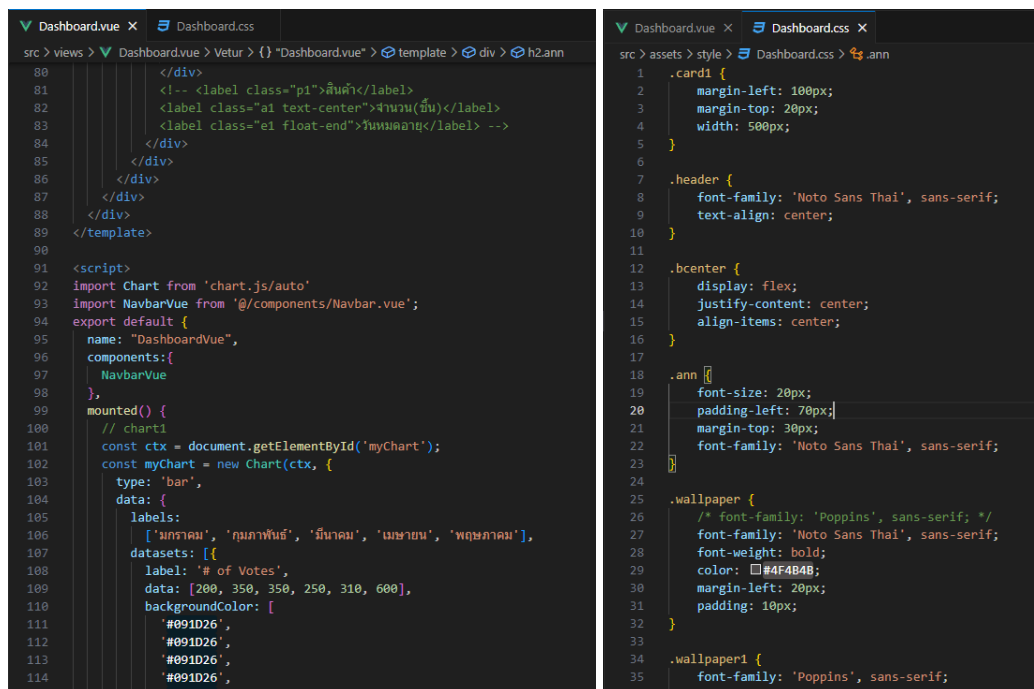


Figure 2.3 Example Vue.js framework^[3]

2.3.4 Figma

Figma is a tool that can be used to design from websites, applications for UX / UI Designers around the world or used for designing logos, various artworks of the Graphic Design field, including people who use in designing Presentations is shown in Figure 2.4

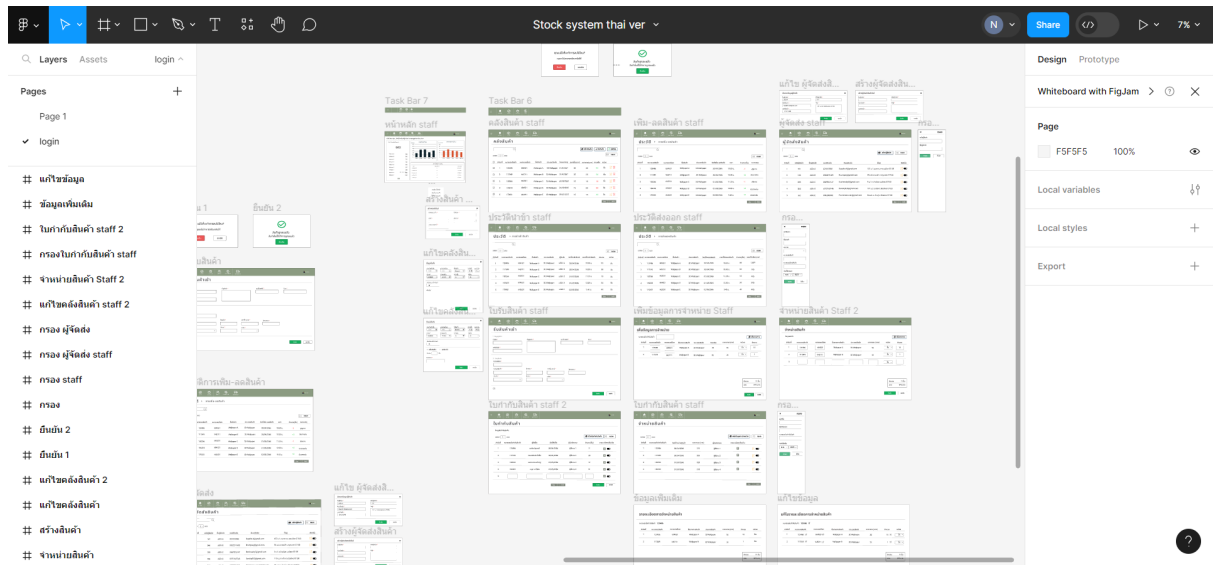


Figure 2.4 Example Figma^[4]

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Workflow

We have simulated how our system works as a workflow in Figure 3.1.

In the system's operational process, users are required to perform a login before accessing its functionalities. If a user is not logged into the system, they will not have the privilege to use its features. The login process is divided into two categories: first, logging in as a regular user, and second, logging in as an administrator. Upon successful login, users will be directed to the main homepage as the default landing page. This differs for administrators, who will have access to a user management menu. Regular users will not see this menu appear in their navigation bar.

Following the homepage, there will be a product management menu where users can choose to either sell products or add products to the inventory. If a user opts to add products, the system will check whether the user has already created that particular product. If not, the user will need to create the product first. Once the product is successfully created, the user can then add it to the system. In the final step, after the user successfully adds a product, the system will update the inventory stock to reflect the addition of the new product. This ensures that the stock levels are accurately maintained in the system, accounting for any changes made through the addition of products. In the sales process, when a user intends to sell a product, the system will verify whether the desired quantity of the product is available in the inventory. If the quantity is insufficient, the user will need to add more of that product to the inventory. If the product quantity is adequate, the user can proceed with the sale, and the system will deduct the sold quantity from the inventory. This ensures that users can only sell products that are currently in stock.

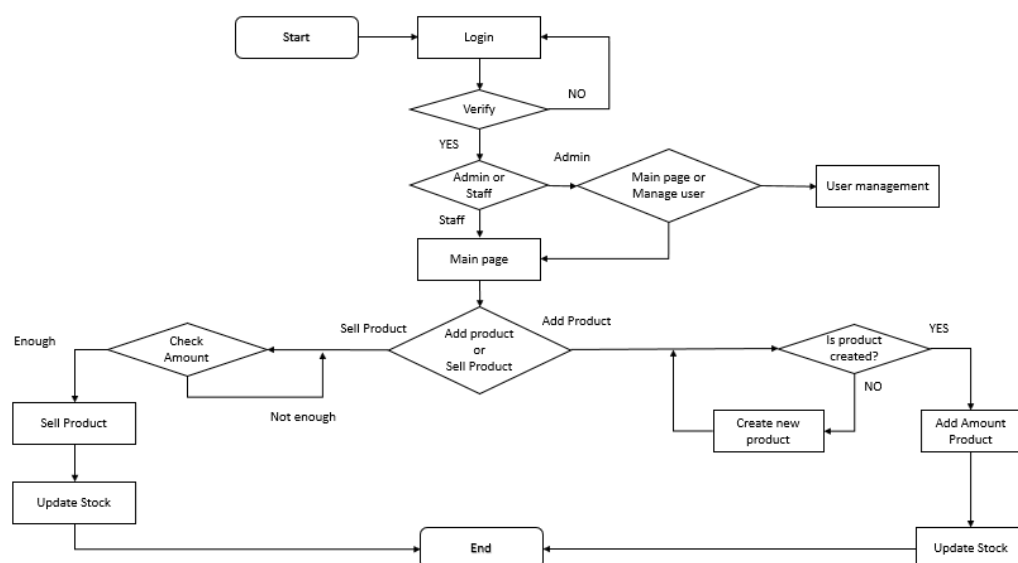


Figure 3.1 Workflow diagram of the system

3.2 Use case diagram

After gathering requirements, the system could be represented by a use case diagram as shown in Figure 3.2. In our stock system, we have 2 roles for users. The first role is admin, admin can use all functionalities such as manage supplier information, manage customers information, etc. In the second role is normal user, the distinction between regular users and administrators lies in the fact that regular users will not have access to user account management. Unlike administrators, who have the capability to manage user accounts, regular users will not be able to perform such actions within the system.

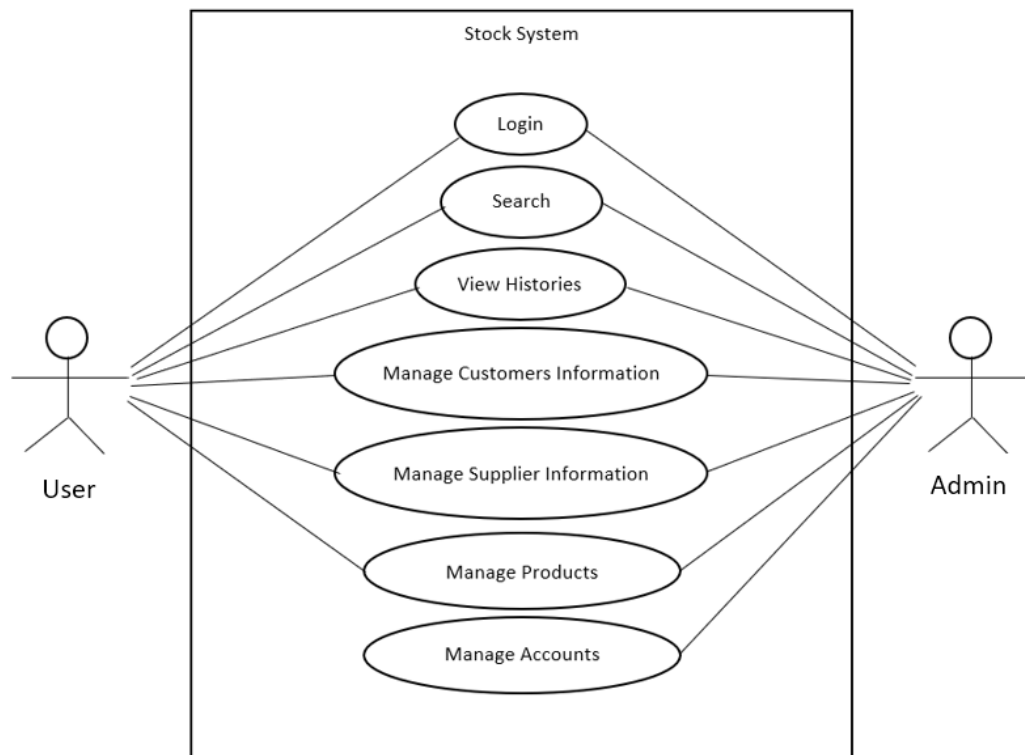


Figure 3.2 Use case diagram of the system

User type	Description
Staff	Staff can Can create, add, update and delete products. Login / Logout system. View inventories. View histories. Can export Excel files. Can edit supplier and customer information.
Admin	Admin can Can create, add, update and delete products. Login / Logout system. View inventories. View histories. Can export Excel files. Can edit supplier and customer information. Can create, update and delete user account

3.3 ER diagram

The system consists of 7 tables, each table has a relationship as shown in the picture.

- 1) Movement Type
This table consists of movement ID and movement name and has a one-to-many relationship to stock history by movement ID is the primary key of this table and foreign key to the stock history table.
- 2) Stock History
This table consists of product lot, product expired date, product time, etc. The stock history table has many relationships to another table such as many-to-one to movement type, many-to-one to product by product lot and product index are the primary key of this table and have many foreign key from the another table.
- 3) Product Unit
This table has unit ID, unit name and has a one-to-many relationship to stock history and also one-to-many relationship to product table. Unit ID is the primary key of this table and foreign key to both stock history table and product table.
- 4) Product
This table has product ID, product name, product type, product sale price and has a one-to-many relationship to stock history, many-to-one relationship to product unit table. Product ID is the primary key in this table, be foreign key to stock history and has unit ID for foreign key from product unit table.

5) Supplier

This table has supplier ID, supplier name, supplier address, supplier telephone number and supplier email. Supplier table has a many-to-one relationship to action type table and one-to-many relationship to stock history table. Supplier ID is the primary key in this table, also be foreign key to the stock history table and has action ID as a foreign key.

6) Action Type

This table has an action ID and action name, Action table has a one-to-many relationship to supplier table and one-to-many relationship to user table. Action ID is the primary key and be foreign key to both the supplier and user table.

7) Users

The users table has user ID, user name, user password, etc. Users table has two relationships between stock history and action type table. By one-to-many relationship and be foreign key to the stock history table. Many-to-one relationship to action type table and has action ID as a foreign key.

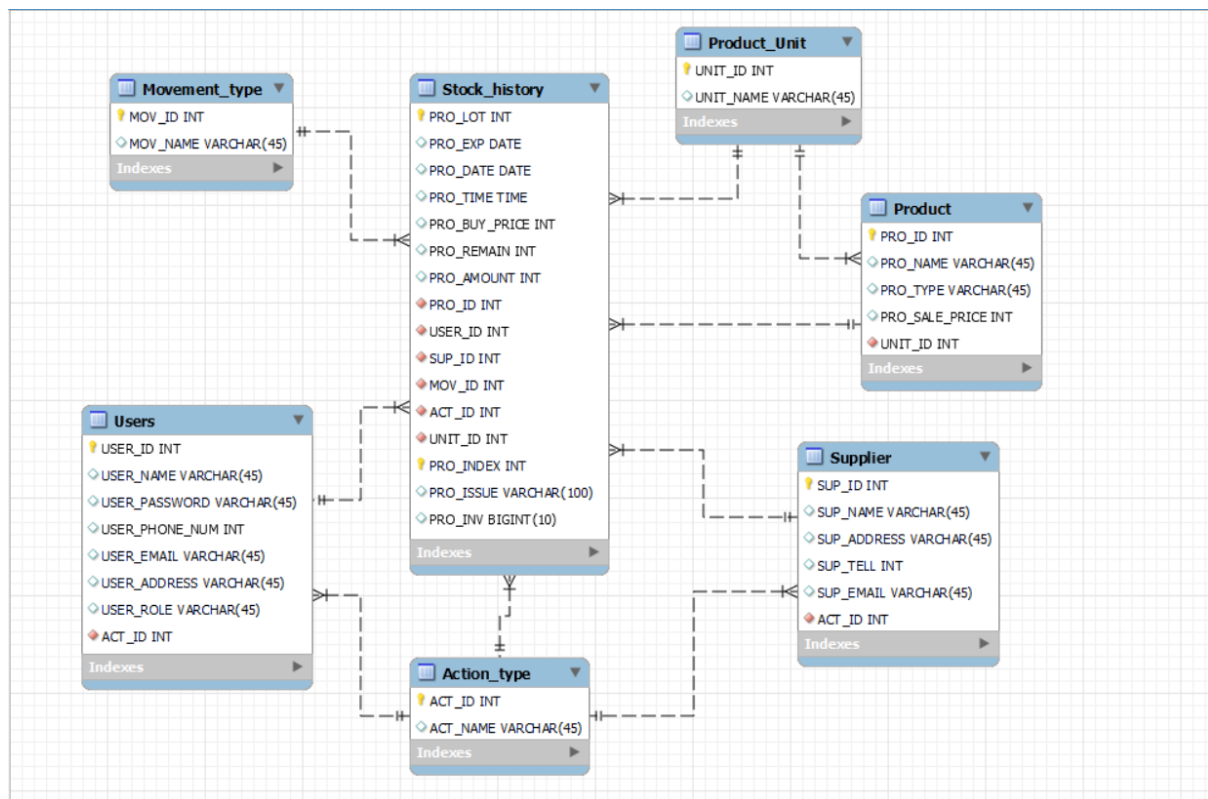


Figure 3.3 ER diagram

Table: User				
Attribute	Description	Data type	Key	Example
USER_ID	User id	Integer	PK	001
USER_NAME	User name	Varchar(45)		Anna Lee
USER_PASSWORD	User password	Integer		001
USER_PHONE_NUMBER	User phone number	Varchar(45)		081222333
USER_EMAIL	User email	Varchar(45)		Anna.lee@gmail.com
USER_ADDRESS	User address	Varchar(45)		333 ม.1 อ.เมือง จ.เชียงราย
USER_ROLE	User role	Varchar(45)		แอดมิน
ACT_ID	User action	Integer	FK	1

Table: Supplier				
Attribute	Description	Data type	Key	Example
SUP_ID	Supplier id	Integer	PK	001
SUP_NAME	Supplier name	Varchar(45)		3d Wallpaper co.
SUP_ADDRESS	Supplier address	Varchar(45)		333 ม.1 อ.เมือง จ.เชียงราย
SUP_TELL	Supplier telephone	Integer		053-333-222-1
SUP_EMAIL	Supplier email	Varchar(45)		3dwallpeper@gmail.com
ACT_ID	Supplier action	Integer	FK	1

Table: Movement_type				
Attribute	Description	Data type	Key	Example
MOV_ID	Movement id	Integer	PK	1
MOV_NAME	Movement name	Varchar(45)		นำเข้า

Table: Product				
Attribute	Description	Data type	Key	Example
PRO_ID	Product id	Integer	PK	001
PRO_NAME	Product name	Varchar(45)		Wood grain wallpaper
PRO_TYPE	Product type	Varchar(45)		3d Wallpaper
PRO_SALE_PRICE	Product sale price	Integer		35
UNIT_ID	Product unit	Integer	FK	1

Table: Product_unit				
Attribute	Description	Data type	Key	Example
UNIT_ID	Unit id	Integer	PK	1
UNIT_NAME	Unit name	Varchar(45)		สี

Table: Action_type				
Attribute	Description	Data type	Key	Example
ACT_ID	Action id	Integer	PK	1
ACT_NAME	Action name	Varchar(45)		ใช้งาน

Table: Stock_history				
----------------------	--	--	--	--

Attribute	Description	Data type	Key	Example
PRO_LOTE	Product lot	Integer	PK	001
PRO_EXP	Product expired date	Date		30/05/2024
PRO_DATE	Product date	Date		18/05/2023
PRO_TIME	Product time	Time		10:00
PRO_BUY_PRICE	Product buy price	Integer		20
PRO_REMAIN	Product remain	Integer		10
PRO_AMOUNT	Product amount	Integer		5
PRO_ID	Product id	Integer	FK	001
USER_ID	User id	Integer	FK	001
SUP_ID	Supplier id	Integer	FK	001
MOV_ID	Movement id	Integer	FK	1
ACT_ID	Action id	Integer	FK	1
UNIT_ID	Unit id	Integer	FK	1
PRO_Index	Product Order Number	Integer	PK	1, 2, 3
PRO_ISSUE	Product Remark	Varchar(100)		Miscount
PRO_INV	Product	Bigint(10)		2077

CHAPTER 4

RESULT

4.1 User Interface

4.1.1 Login

This is a Login page where users can login to the system by input user ID and password.

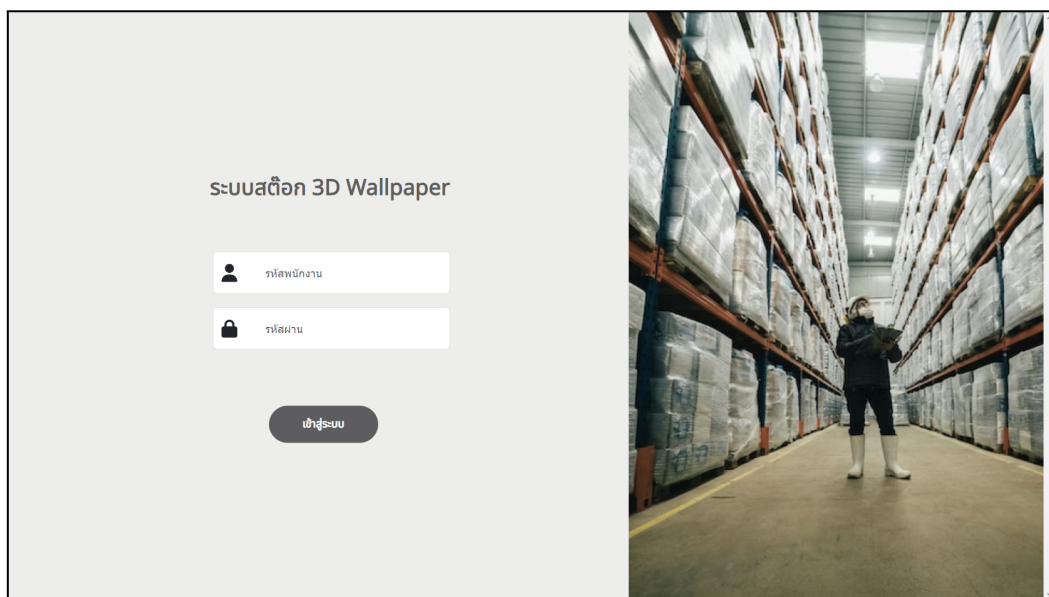


Figure 4.1 Login page

4.1.2 Website menus

This is the menu bar of the website that users can choose to use such as maintain inventory, history, receive products, users, sell, supplier and create products.

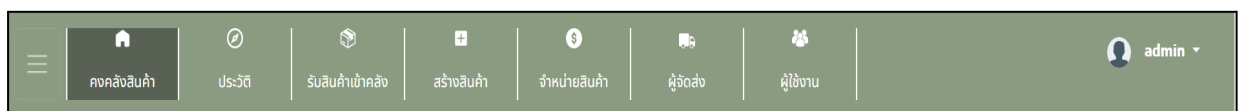


Figure 4.2 Website menus bar

4.1.3 Maintain Inventory

This Maintain Inventory page will show a detailed and information summary of the system such as show amount products in stock, monthly sales, monthly income and expiration date.

ลำดับที่	หมายเลขสินค้า	ชื่อสินค้า	ประเภทสินค้า	ราคาขาย(บาท)	คงเหลือ	หน่วย
1	1	เครื่องปั่น	อุปกรณ์อิเล็กทรอนิกส์	120000	3	เครื่อง
2	101101	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	12000	60	เครื่อง
3	101102	เบาสี	อุปกรณ์อิเล็กทรอนิกส์	120	10	ชิ้น
4	101103	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	300	30	ชิ้น

Figure 4.3 Maintain Inventory page

4.1.4 Main receive products into warehouse

The Main receive products page will show all inventory products that users have added into inventory. In this page, users can import product, edit, update product and also export product information to excel file.

ลำดับที่	หมายเลขสินค้า	หมายเลขซื้อ	ชื่อสินค้า	ประเภทสินค้า	วันหมดอายุ	ราคาซื้อ(บาท)	ราคาขาย(บาท)	คงเหลือ	หน่วย	other
1	1	101111	เครื่องปั่น	อุปกรณ์อิเล็กทรอนิกส์	2023/10/30	5000	120000	3	เครื่อง	✎ ✖
2	101101	101105	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	2023/11/08	10000	12000	40	เครื่อง	✎ ✖
3	101101	101109	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	2563/05/22	100	12000	20	เครื่อง	✎ ✖
4	101102	101102	เบาสี	อุปกรณ์อิเล็กทรอนิกส์	2023/10/31	100	120	10	ชิ้น	✎ ✖
5	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	2023/11/09	500	300	30	ชิ้น	✎ ✖

Figure 4.4 Main receive products page

4.1.5 Receive products into warehouse

Receive products page will show all inventory products information and have several product details such as supplier ID, product LOT, product ID etc. You must fill out all information in order to add product information to your inventory.

Figure 4.5 Receive products page

4.1.6 Create new product

In menu bar, you can create new product as you want by pressing “สร้างสินค้า” button. After that, you fill out the product details completely.

ลำดับที่	หมายเลขสินค้า	ชื่อสินค้า	ประเภทสินค้า	หน่วย	ราคาขาย (บาท)
1	1	เครื่องปรีน	อุปกรณ์อิเล็กทรอนิกส์	เครื่อง	120000
2	101101	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	เครื่อง	12000
3	101102	เมาส์	อุปกรณ์อิเล็กทรอนิกส์	ชิ้น	120
4	101103	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	ชิ้น	300
5	101104	แผ่นรองเมาส์	อุปกรณ์เสริม	ชิ้น	100

Figure 4.6 Create New Product Page

4.1.7 Product information

In the inventory page, if you want to view the information or want to update and edit the information, you can do this by pressing the pen icon.

Figure 4.7 Product information

4.1.8 Search function

This function allows you to search products as you like by type the product ID in the search field.

รับสินค้าเข้าคลังสินค้า

5

🔍

→ รับสินค้า

📄

EXCEL

แสดง 5

แถว

ลำดับที่	หมายเลขสินค้า	หมายเลขจัด	ชื่อสินค้า	ประเภทสินค้า	วันหมดอายุ	ราคาซื้อ(บาท)	ราคาขาย(บาท)	คงเหลือ	หน่วย	other
1	101105	101101	แผ่นรองแก้ว	ของใช้	2023/11/11	20	50	20	ชุด	☑️ 🗑️
2	101105	101103	แผ่นรองแก้ว	ของใช้	2023/10/31	15	50	50	ชุด	☑️ 🗑️
3	101105	101102	แผ่นรองแก้ว	ของใช้	2023/11/01	20	50	10	ชุด	☑️ 🗑️

ย้อนกลับ

1

ถัดไป

Figure 4.8 Inventories information filtering

4.1.9 Confirm delete

This function will alert you if you press the delete button. and ask if you are sure to delete this product.



Figure 4.9 Confirm delete first step

4.1.10 Confirm after delete

After you are sure to delete the information of that product. The system will confirm again that You have successfully deleted that product. Then that item will not appear in the inventory page.

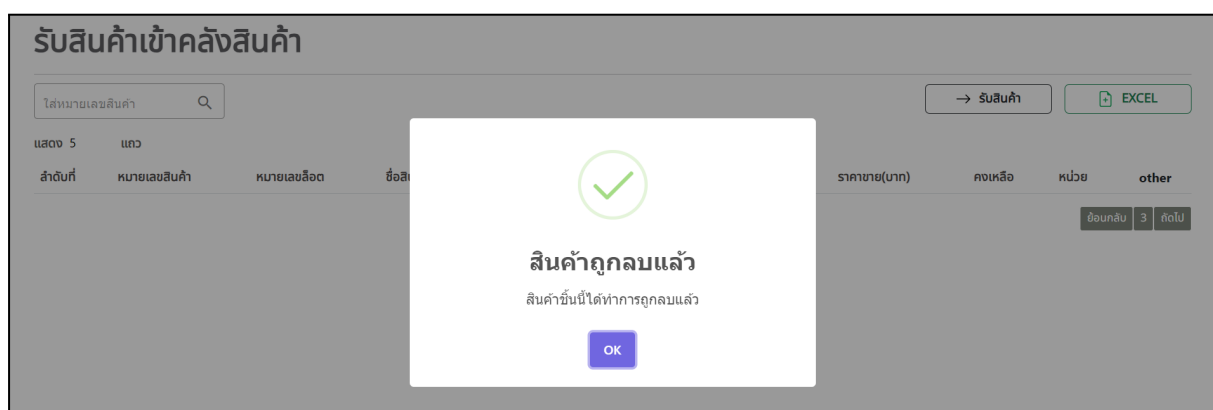


Figure 4.10 Confirm Delete Successful

4.1.11 View histories options

These options will be an option that will let you choose which part of the product history do you want to see first? We will have 3 types of history pages to choose from: Import-products histories, export products histories and increase-decrease products histories.

ลำดับที่	หมายเลขสินค้า	หมายเลขล็อต	ชื่อสินค้า	ประเภทสินค้า	ผู้จัดส่ง	วันที่นำเข้าสินค้า	เวลาที่นำเข้าสินค้า	จำนวน	หน่วย
1	101105	1011	แผ่นรองแก้ว	ของใช้	Pepse	2023/12/08	02:37:54	5	ชุด
2	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/11/20	02:45:37	4	ชิ้น
3	101101	101109	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	Pepse	2023/10/24	23:21:30	1	เครื่อง
4	101102	101102	เมาส์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/10/18	03:40:54	12	ชิ้น
5	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/10/18	03:40:18	30	ชิ้น

Figure 4.11 3 Options of view histories

4.1.12 Import products histories

This page shows the histories of all imported products. And you can also filter to find what you're looking for.

ลำดับที่	หมายเลขสินค้า	หมายเลขล็อต	ชื่อสินค้า	ประเภทสินค้า	ผู้จัดส่ง	วันที่นำเข้าสินค้า	เวลาที่นำเข้าสินค้า	จำนวน	หน่วย
1	101105	1011	แผ่นรองแก้ว	ของใช้	Pepse	2023/12/08	02:37:54	5	ชุด
2	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/11/20	02:45:37	4	ชิ้น
3	101101	101109	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	Pepse	2023/10/24	23:21:30	1	เครื่อง
4	101102	101102	เมาส์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/10/18	03:40:54	12	ชิ้น
5	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/10/18	03:40:18	30	ชิ้น

Figure 4.12 Histories of import products page

4.1.13 Export products histories

This page is the same as import products histories but this page will show the histories of all export products. And you can also filter to find what you're looking for.

ลำดับที่	หมายเลขสินค้า	หมายเลขล็อต	ชื่อสินค้า	ประเภทสินค้า	วันที่ส่งออกสินค้า	เวลาที่ส่งออกสินค้า	จำนวน	หน่วย	รวมเป็นเงิน(บาท)
1	1	101111	เครื่องปั่น	อุปกรณ์อิเล็กทรอนิกส์	2023/12/08	13:26:07	7	เครื่อง	840000

Figure 4.13 Histories of export products page

4.1.14 Increase – Decrease products histories

The last one, this page will show you all about increase and decrease information of products histories. And show the note or cause of the increase or decrease of the product.

ลำดับที่	หมายเลขสินค้า	หมายเลขล็อต	ชื่อสินค้า	ประเภทสินค้า	วันที่แก้ไข	เวลาที่แก้ไข	จำนวน	ผู้ใช้งาน	หมายเหตุ
1	1	101111	เครื่องปั่น	อุปกรณ์อิเล็กทรอนิกส์	2023/12/08	13:35:29	+5	admin	ซื้อ
2	101103	101103	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	2023/12/04	14:55:50	+10	admin	ตกหล่น
3	101103	101103	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	2023/12/04	14:55:18	-10	admin	นับผิด
4	1	101111	เครื่องปั่น	อุปกรณ์อิเล็กทรอนิกส์	2023/11/28	13:56:57	-5	admin	หาย
5	101101	101105	คอมพิวเตอร์	อุปกรณ์อิเล็กทรอนิกส์	2023/11/28	13:53:56	-5	admin	หาย

Figure 4.14 Histories Increase – Decrease products page

4.1.15 Search Histories

This function you can search the history of product information as you like by filling in the product_ID , then pressing the ENTER button of your keyboard. After that, product history will show in the history page. You can search by product ID.

ประวัติ ▶ การนำเข้าสินค้า									
3 Q									
แสดง 5 แถว									
ลำดับที่	หมายเลขสินค้า	หมายเลขล็อต	ชื่อสินค้า	ประเภทสินค้า	ผู้จัดส่ง	วันที่นำเข้าสินค้า	เวลาที่นำเข้าสินค้า	จำนวน	หน่วย
1	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/11/20	02:45:37	4	ชิ้น
2	101103	101101	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/10/18	03:40:18	30	ชิ้น
3	101103	101103	แป้นพิมพ์	อุปกรณ์อิเล็กทรอนิกส์	Nvidia	2023/10/18	02:15:08	20	ชิ้น
								ย้อนกลับ	1 ถัดไป

Figure 4.15 Search Histories

4.1.16 Supplier

This supplier page will show you information and details of suppliers such as supplier ID, supplier name, supplier telephone number, etc. Also, you can create new supplier information and search suppliers by search bar.

คงคลังสินค้า ประวัติ รับสินค้าเข้าคลัง สร้างสินค้า จำหน่ายสินค้า ผู้จัดส่ง ผู้ใช้งาน admin ▾						
ผู้จัดส่งสินค้า						
กรองกรหัสผู้จัดส่ง Q สร้างผู้จัดส่ง						
แสดง 5 แถว						
ลำดับที่	รหัสผู้จัดส่ง	ชื่อผู้จัดส่ง	เบอร์ติดต่อ	อีเมลติดต่อ	ที่อยู่	ฟังก์ชัน
1	13	Pepse	05315616462	Pepse@gmail.com	364/44 กรุงเทพมหานคร	☒ ☐
2	101101	Nvidia	05333333332	Nvidia@gmail.com	NVidia Corporation	☒ ☐
						ย้อนกลับ 1 ถัดไป

Figure 4.16 Supplier Information

4.1.17 Create new supplier

If you want to add or create a new supplier, you can press the “สร้างผู้จัดส่ง” button to create a new supplier's information.

Figure 4.17 Create new supplier information

4.1.18 Update & edit supplier information

This function you can update or edit supplier information as you like by pressing the pen icon. On this page, all suppliers' information is displayed.

Figure 4.18 Update or edit supplier information

4.1.19 Supplier Searching

This function you can search the supplier information as you like by filling in the search bar and using supplier ID. You can search by supplier ID.

Figure 4.19 Supplier Searching

4.1.20 Sale page

This page is about exporting the product out of the stock when selling, where this page will show the number of invoices, date, total price and name of the user that used at that time. There are 2 functions on it such as 1. Add product information, 2. More detailed function.

Figure 4.20 Sale page

4.1.21 Add product information

This page is from the sale page through the “เพิ่มข้อมูลการจำหน่าย”. This page is about keeping a list of sold products in invoice for updating the remainder in the stock. The system will let user fill invoice number and fill the product number then the information will show up user can fill product amount if everything is correct, click “เพิ่มข้อมูลการจำหน่าย”.

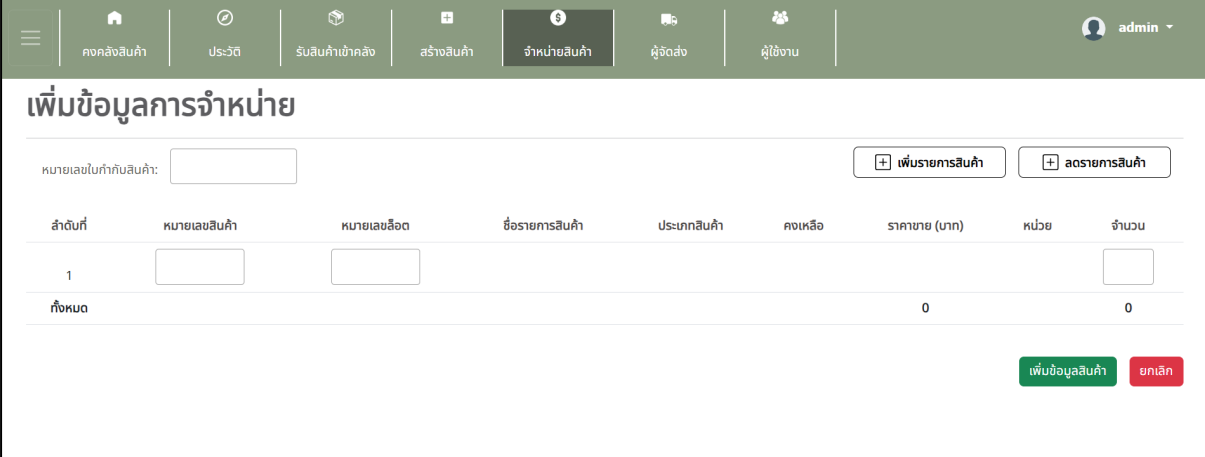
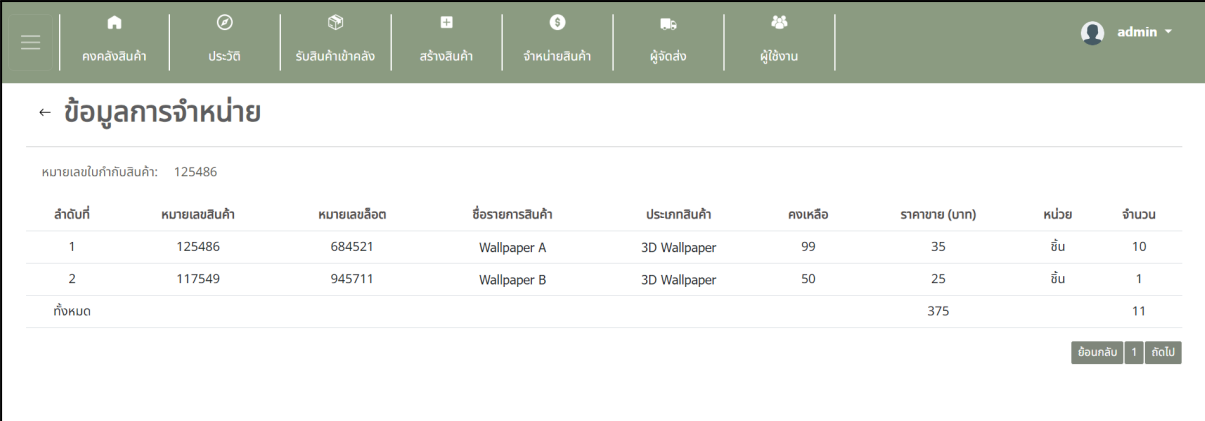


Figure 4.21 Add product information

4.1.22 Selling product information

In this function, users will be able to check what products were sold in one sale or one invoice. This function also displays product list information, total number of products and total price of products sold.

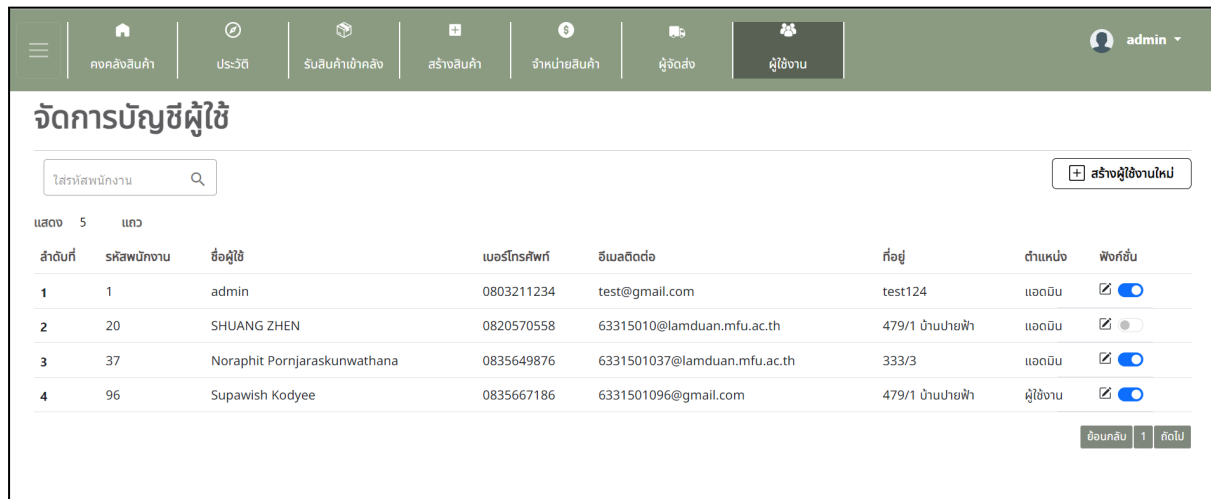


ลำดับที่	หมายเลขสินค้า	หมายเลขซื้อ	ชื่อรายการสินค้า	ประเภทสินค้า	คงเหลือ	ราคาขาย (บาท)	หน่วย	จำนวน
1	125486	684521	Wallpaper A	3D Wallpaper	99	35	ชิ้น	10
2	117549	945711	Wallpaper B	3D Wallpaper	50	25	ชิ้น	1
ทั้งหมด						375		11

Figure 4.22 Selling product information

4.1.23 User management

In part of this page only the admin has the right to access the information of the user. Normal users who are not admins do not have access to the user management page. This page will show all user information displayed.



ลำดับที่	รหัสพนักงาน	ชื่อผู้ใช้	เบอร์โทรศัพท์	อีเมลติดต่อ	ที่อยู่	ตำแหน่ง	พังก์ชั่น
1	1	admin	0803211234	test@gmail.com	test124	แอดมิน	☒
2	20	SHUANG ZHEN	0820570558	63315010@lamduan.mfu.ac.th	479/1 บ้านปายฟ้า	แอดมิน	☐
3	37	Noraphit Pornjaraskunwathana	0835649876	6331501037@lamduan.mfu.ac.th	333/3	แอดมิน	☒
4	96	Supawish Kodyee	0835667186	6331501096@gmail.com	479/1 บ้านปายฟ้า	ผู้ใช้งาน	☒

Figure 4.23 User Management page

4.1.24 Create new user

This page, admin can create a new user by pressing “สร้างผู้ใช้งานใหม่” button and then complete the user's details along with specifying the user's role as an administrator or general users.

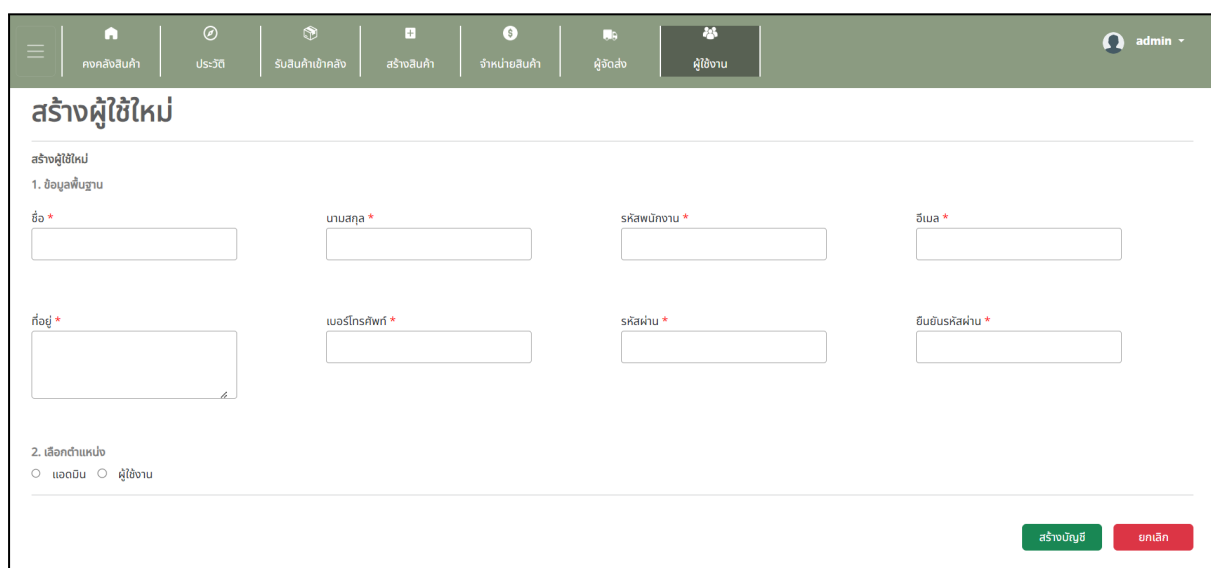


Figure 4.24 Create new user page

4.1.25 Update & edit user information

In part of this page only the admin has the right to access the information of the user.

Admin will be able to update, edit or change the information of all users by pressing the pen icon.

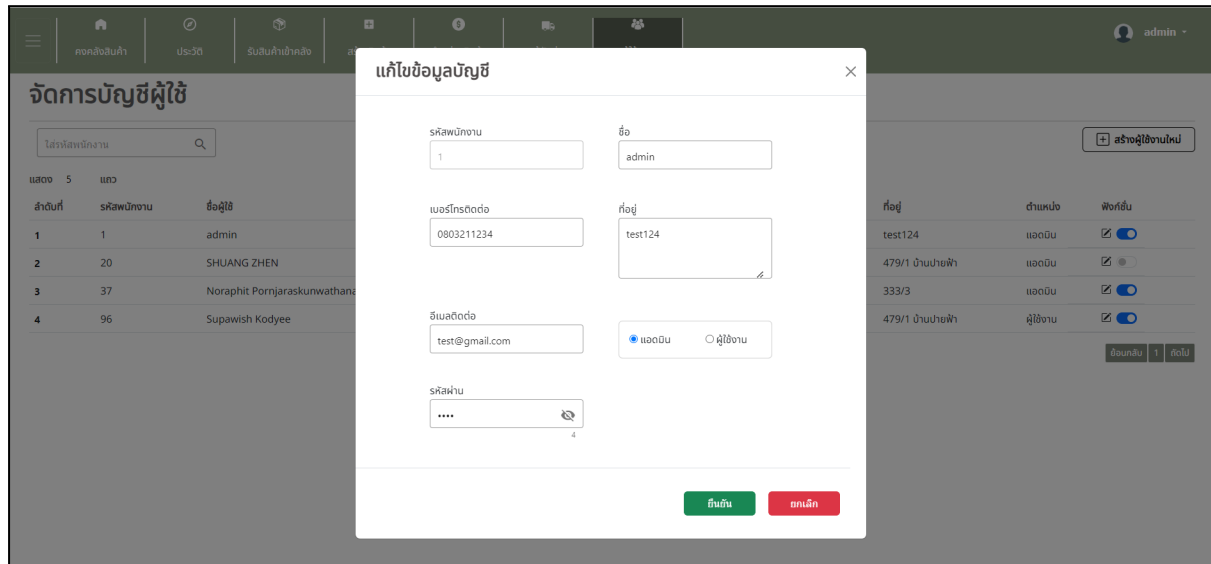


Figure 4.25 Update & edit user information

4.1.26 User Permissions function

In this function, only users with admin status have the right to grant users whether users have the right to access the system. If this function is on (switch on) means that the user can login into the system, if this function is off (switch off) means that the user can't login into the system or doesn't have permission to login to the system.

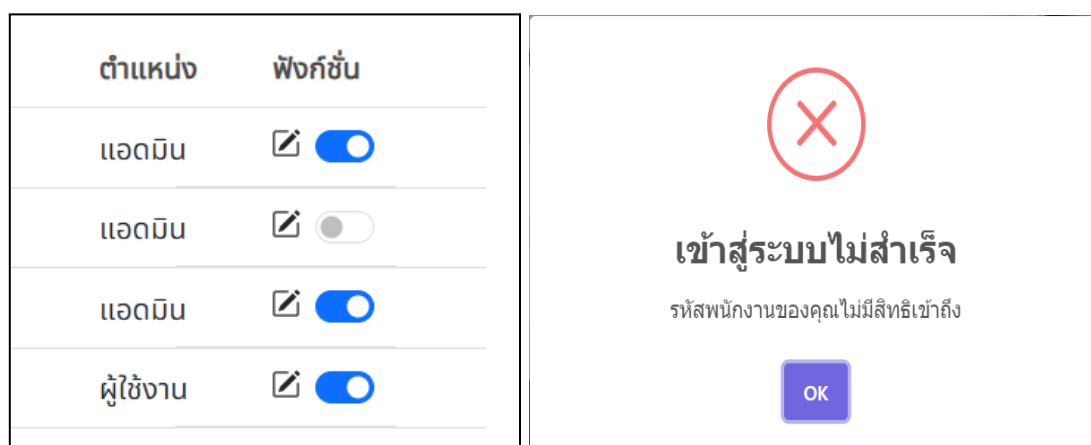


Figure 4.26 User permissions

4.1.27 Logout system

If you want to log out, you can do this by pressing the profile button and then press on logout. After you press logout. The system will take you back to the login page again.

The screenshot shows a web application interface with a top navigation bar and a main content area. The top navigation bar includes a menu icon, a home icon, a clock icon, a document icon, a plus icon, a dollar sign icon, a mail icon, and a profile icon. The profile icon is highlighted, and a dropdown menu is visible with the text 'admin' and a red button labeled 'ออกจากระบบ' (Logout). The main content area has a title 'จัดการบัญชีผู้ใช้' (Manage User Accounts) and a search bar labeled 'ใส่รหัสพนักงาน' (Enter Employee ID). Below the search bar is a table with 4 rows and 8 columns. The columns are: ลำดับที่ (Serial Number), รหัสพนักงาน (Employee ID), ชื่อผู้ใช้ (Username), เบอร์โทรศัพท์ (Phone Number), อีเมลติดต่อ (Contact Email), ที่อยู่ (Address), ตำแหน่ง (Position), and ฟังก์ชัน (Function). The table contains 4 rows of data. At the bottom right of the table, there are buttons labeled 'ย้อนกลับ' (Go Back), '1', and 'ถัดไป' (Next).

ลำดับที่	รหัสพนักงาน	ชื่อผู้ใช้	เบอร์โทรศัพท์	อีเมลติดต่อ	ที่อยู่	ตำแหน่ง	ฟังก์ชัน
1	1	admin	0803211234	test@gmail.com	test124	แอดมิน	☒
2	20	SHUANG ZHEN	0820570558	63315010@lamduan.mfu.ac.th	479/1 บ้านป่ายฟ้า	แอดมิน	☐
3	37	Noraphit Pornjaraskunwathana	0835649876	6331501037@lamduan.mfu.ac.th	333/3	แอดมิน	☒
4	96	Supawish Kodyee	0835667186	6331501096@gmail.com	479/1 บ้านป่ายฟ้า	ผู้ใช้งาน	☒

Figure 4.27 Logout system

CHAPTER 5

CONCLUSION

5.1 Conclusion

We have researched many examples and information to create, design and develop our stock management system. We have set the scope of web development from research planning and found the reference from system design to make our web application work and more efficient. Throughout the project period from the planning and design, starting with design UX/UI creation by using Figma. In terms of system development, it is an important work step for a successful project. Then, we designed, planned and assigned tasks of system diagram, use-case diagram and workflow. After that, the system will then continue to be improved so that it is as helpful to everyone as possible.

5.2 Discussion

This project has received advice and guidance from many Professors to create, design and develop the system. We learned and tried to understand the stock management system. This project had some difficulties in working and planning in the part that we have little software knowledge and stock management knowledge, which the knowledge we have is not enough to proceed. This is a very highly challenging project because we must invest time in learning, practicing, and researching stock management, system design, learning programming language and operation. However, the project is moving along without any issues and waiting for the web building to develop and continue currently. Now, we are learning and studying how to design the web page of the system by coding for the next step.

In the future, we may have various features added to the system that can enhance convenience and speed up the workflow for users. Examples include scanning barcodes for importing and selling products, receiving notifications for product expiration dates, and alerts for products that are about to run out. The system may also display income reports from product sales. All of these are features that we anticipate in the future.

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